CECS 229 – Assignment 9

Gaussian Elimination

The file, 'pa9.py', contains an incomplete set of functions that will transform a matrix into Row Echelon Form.

Update the file 'pa8.py' that you created in the previous assignment to add the following new methods to the Matrix class that provide functionality to perform the transformations to REF:

1. `copy(self)` – this method returns a distinct copy of the matrix that will be used to perform the operations on.
    a. Construct and return a new Matrix object with the same dimensions that has a copy of all of the elements in the same locations.
2. `swap_rows(self, i, j)` – this method swaps row i with row j within the matrix. No value is returned, the matrix itself is updated.
    a. i – the index of the first row to be swapped (1 based).
    b. j – the index of the second row to be swapped (1 based).
3. `scale_row(self, i, c)` – this method multiplies all elements in row i by the scalar c. No value is returned, the matrix itself is updated.
    a. i – the index of the row (1 based).
    b. c – the scalar to multiply the elements by.
4. `add_row_multiple(self, src, tar, c)` – this method multiplies the source row by the scalar value, adds the result to the target row and then overwrites the target row with that result. No value is returned, the matrix itself is updated.
    a. src – the index of the source row (1 based).
    b. tar – the index of the target row (1 based).
    c. c – the scalar to multiply the elements by.

Update the file 'pa9.py' that you created in the previous assignment to add the following new methods to the Matrix class that provide functionality to perform the transformation to REF:

1. `pivot_index(i, j, A)` – locates the pivot position of Matrix A at or after location i, j and returns a tuple of the pivot location (row, col) using 1 based indexing. If no pivot exists, then return None.
    a. i – the index of the row where the search for the pivot will begin (1 based).
    b. j – the index of the column where the search for the pivot will begin (1 based).
    c. A – the matrix where the pivot will be searched for.
    d. Returns a tuple (row, col) of the location of the pivot that was at or after location i, j, or returns None if no pivot exists.
2. `ref(A)` – returns a new Matrix object of Matrix A transformed into row echelon form.
    a. A – the matrix that will be transformed in
    b. Returns a Matrix object with the elements in REF format.
    c. Note: this function should make a copy of A and perform the operations on the copy rather than modifying matrix A.

Notes:

1. Please place your name in a comment block at the top of both ‘pa8.py’ and ‘pa9.py’.
2. Add the specified methods to the ‘pa8.py’ file that you wrote in the previous assignment.
3. Fill in the necessary code to complete the functions in ‘pa9.py’. Write your code where specified (ie. the TODOs). Do not modify the file names, function names, method names, or parameters otherwise the program and tests will not work. Return the results in the format specified (ie. correct types and container types), otherwise the program and tests will not work properly.
4. Your solutions may not use any libraries or functions that perform the equivalent task.
5. Test your code using the provided ‘main.py’ file. You can write additional tests in the ‘your_tester.py’ file to further check your solutions. Note: just because the main tests pass does not guarantee that it will pass the tests on Gradescope.
6. When you are finished, submit both your ‘pa8.py’ and ‘pa9.py’ files to Gradescope on Canvas for further testing and grading. It should allow you to make repeated submissions.